Assignment #4. Forms and Validation (HTML + Bootstrap)

Objective:

By the end of this assignment, students will be able to:

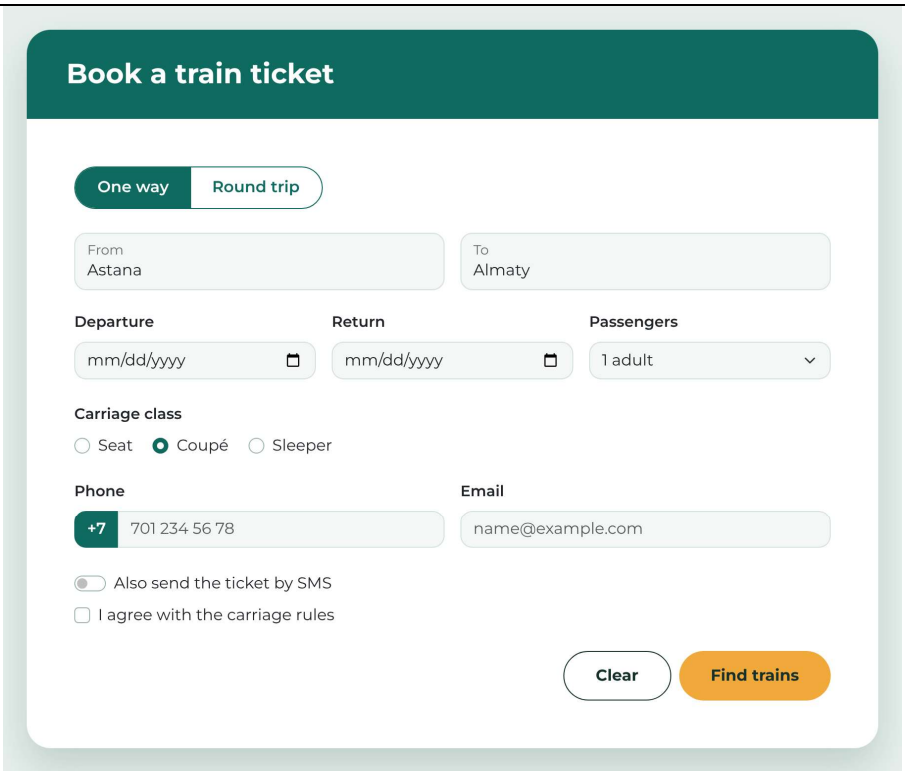
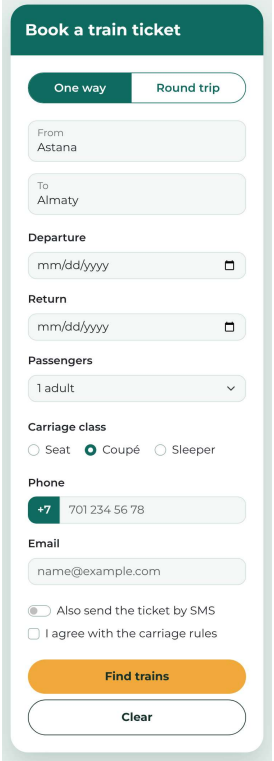
- build forms with Bootstrap form components and the grid;
- validate a form with native HTML attributes and show clear error messages;
- find and fix the usability and accessibility problems of a form.

Rules for Submission:

1. This is an individual assignment. If the works of different students are the same, the grade is **0**.
2. Submit before the deadline. **Late works are not accepted!** To get a grade you must **submit before the deadline AND defend your work at the practice lesson**. Submission **without a report** or **without a defense** will result in a grade of 0.
3. You must submit: your report in **Assignment4_YourName.docx** (or .pdf) format. If you submit in other formats, your grade will be **0**. All source code must be uploaded to the GitHub repository.
4. If you cannot explain your own code, the whole assignment is graded 0. At the defense your work will be tested with new values that are not in the tables.
5. No JavaScript and no novalidate attribute in this assignment.
6. IN THE REPORT MUST BE:
   - Your name and group
   - Every task with screenshots and a short explanation in your own words
   - Task 1 at 375px and 1280px next to the picture; Task 2: the table with your results; Task 3: the table of problems.
   - A short conclusion about what you learned
   - The **GitHub repository URL** with the source code. Submissions without the link will receive a grade of 0

Task 1. Book a train ticket (Bootstrap forms + your CSS)

Reproduce this design. Almost every row is a different Bootstrap form component: recognize it, find it in the documentation and then restyle it with your own CSS on top of Bootstrap: colors, the Montserrat font, rounded corners, the shadow and the buttons. Sizes may differ slightly.

Laptop (1280px)	Phone (375px)
	

Colors: #0F6B5F, #F2A93B, #16302B, #F4F7F5, #E6EFEC.

Please note:

- **"One way / Round trip"** is two radio inputs; on a phone **"Find trains"** is above **"Clear"** without changing the HTML order;
- **From, To, Departure, Carriage class, Phone, Email** and **the agreement** are required; **Passengers** is 1–6 adults; **"Clear"** empties the form;
- every field has a connected label, and the form works with the keyboard only.

Tools: Bootstrap grid, form and utility classes and your own CSS. No fixed widths and no absolute positioning.

Learn more: https://getbootstrap.com/docs/5.3/forms/overview/

Task 2. Registration for AITU Hack 2026 (HTML validation)

Build the registration form of a student hackathon with Bootstrap. Write every rule with HTML attributes only, without the pattern attribute: choose the right input type and its attributes, so that the browser itself refuses bad data. Every value in the table must give the shown result.

Fields marked with * are required.

Field	Rule	Must accept	Must reject
Full name *	2–40 characters	Anna-Maria Lee	A
University email *	a valid email address	a.fedenko@astanait.edu.kz	a.fedenko a.fedenko@
Age *	a whole number from 16 to 30	18	15 31
Team size *	a whole number from 1 to 4	4	5 2.5
Track *	Web, Mobile, AI or GameDev	Web	“Choose…”
Arrival day *	20–22 November 2026	21.11.2026	25.11.2026
Check-in time *	09:00–18:00, every 30 minutes	10:30	10:15 19:00
Entry fee *	0–5000 ₸, in steps of 500 ₸	1500	1250 6000
Experience level *	one of three radio buttons	Beginner	nothing chosen
GitHub profile	optional, a full link	(empty) https://github.com/yerassyl-dev	github.com/yerassyl-dev
Password *	8–20 characters	Astana2026	short
Motivation *	50–300 characters	120 characters	49 characters
Rules agreement *	checkbox	ticked	not ticked

Note: Errors (red border and the **.invalid-feedback** text) appear only after the user has changed a field, not when the page opens.

Hints:

- Bootstrap hides **.invalid-feedback** by default. The CSS pseudo-classes **:user-invalid** and **:user-valid** work only after the user has changed a field, for example **input:user-invalid ~ .invalid-feedback { display: block; }**.
- Every rule in the table can be written with **type, required, min, max, step, minlength** and **maxlength**. Choose the input type first: it already checks a lot.
- step works not only with numbers: check what it means for date and time inputs.

Learn more: https://www.w3schools.com/html/html_form_attributes.asp

https://developer.mozilla.org/en-US/docs/Web/HTML/Element/input/time

https://developer.mozilla.org/en-US/docs/Web/CSS/:user-invalid

Task 3. Fix a bad form (HTML + Bootstrap)

A café published this table reservation form. It looks almost normal, but it is hard to use on a phone, with the keyboard or with a screen reader, and part of the data is never sent.

```
<h2>Reserve a table</h2>
<form>
  <input type="text" placeholder="Your name">
  <input type="text" placeholder="Email">
  <label>Phone</label>
  <input type="text" id="phone">
  <label for="guests">Guests</label>
  <input type="text" id="people">
  <label for="date">Date</label>
  <input type="text" id="date" placeholder="dd/mm/yyyy">
  <p>Seating:</p>
  <input type="radio" name="inside" id="date"> Inside
  <input type="radio" name="terrace"> Terrace
  <select>
    <option>Lunch</option>
    <option>Dinner</option>
  </select>
  <textarea placeholder="Comments"></textarea>
  <input type="checkbox"> I agree to the terms
  <div class="btn btn-primary">Reserve</div>
</form>
```

- Find at least 10 problems and write them in a table: the line of code, why it is a problem for the user, how you fixed it. For 2 of the problems, give one more possible fix and explain why you chose yours.
- Rebuild the form in **task3.html** with Bootstrap: the same fields, no errors in the W3C Markup Validator, and it works with the keyboard only.

Assessment criteria

Criterion	What is checked	Points
Task 1	- matches the picture at 375px and 1280px: components, colors, font, corners, buttons - everything from “Please note” works	**15**
Task 2	- the table values and new values from the instructor give the right result - the rules are written with HTML attributes only; errors appear after the user changes a field	**15**
Task 3	- at least 10 real problems, each explained from the user’s side; another possible fix for 2 of them - the fixed form keeps all fields, passes the W3C validator, works with the keyboard	**15**
Report	- contains everything listed in the rules; screenshots can be read; explanations in the student’s own words	**5**
Defense	- explains the work and demonstrates it live without errors (10) - answers the instructor’s questions about their own code and the topic (15) - completes the live coding task independently, without assistance or hints (25)	**50**
Total		**100**

Good luck!